Chapter 4
The Relational Model

Objectives

- Terminologies of relational model.
- How tables are used to represent data.
- Properties of database relations.
- Candidate key, Primary key and Foreign Key.
- Meaning of entity integrity and referential integrity.
- Why do NoSQL systems emerge in the 2000s?

Relational Model Concepts

- The relational model was first proposed by Dr. T.F. Codd of IBM in 1970 in the following paper:

  "A Relational Model for Large Shared Data Banks," Communications of the ACM, June 1970.

- The above paper caused a major revolution in the field of Database management and earned Ted Codd the ACM Turing Award.

Relational Model Terminologies

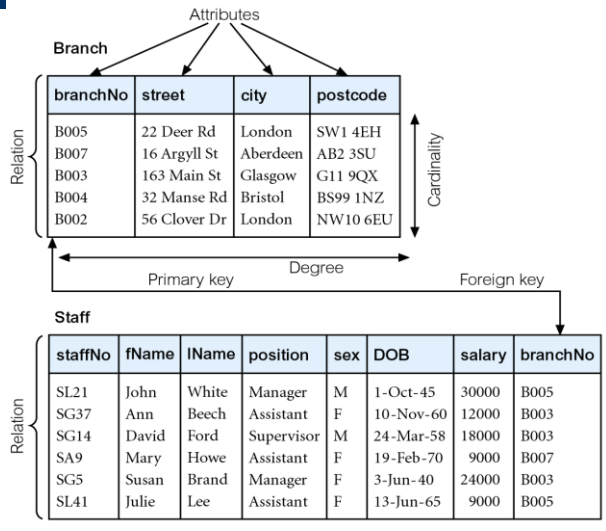
- Data is organized into tables (relations), with columns (attributes) and rows (tuples)
  - Only applies to logical structure of the database, not the physical structure.
- Attribute is a named column of a relation.

  Attribute types:
  - Simple and composite attributes
  - Single-valued and multi-valued attributes
  - Null attributes
  - Derived attributes

Relational Model Terminologies

- Domain defines the **data type and the set of allowable values for one or more attributes.**
  - E.g. Dates have various formats such as yyyy-mm-dd, or dd mm,yyyy etc.
- Tuple is a row / record of a relation.
- Degree is the number of attributes in a relation.
- Cardinality is the number of tuples in a relation.

Instances of Branch and Staff Relations (Fig. 4.1)



Branch

branchNo	street	city	postcode
B005	22 Deer Rd	London	SW1 4EH
B007	16 Argyll St	Aberdeen	AB2 3SU
B003	163 Main St	Glasgow	G11 9QX
B004	32 Manse Rd	Bristol	BS99 1NZ
B002	56 Clover Dr	London	NW10 6EU

Staff

staffNo	fName	lName	position	sex	DOB	salary	branchNo
SL21	John	White	Manager	M	1-Oct-45	30000	B005
SG37	Ann	Beech	Assistant	F	10-Nov-60	12000	B003
SG14	David	Ford	Supervisor	M	24-Mar-58	18000	B003
SA9	Mary	Howe	Assistant	F	19-Feb-70	9000	B007
SG5	Susan	Brand	Manager	F	3-Jun-40	24000	B003
SL41	Julie	Lee	Assistant	F	13-Jun-65	9000	B005

Examples of Attribute Domains (Fig. 4.2)

Attribute	Domain Name	Meaning	Domain Definition
branchNo	BranchNumbers	The set of all possible branch numbers	character: size 4, range B001–B999
street	StreetNames	The set of all street names in Britain	character: size 25
city	CityNames	The set of all city names in Britain	character: size 15
postcode	Postcodes	The set of all postcodes in Britain	character: size 8
sex	Sex	The sex of a person	character: size 1, value M or F
DOB	DatesOfBirth	Possible values of staff birth dates	date, range from 1-Jan-20, format dd-mmm-yy
salary	Salaries	Possible values of staff salaries	monetary: 7 digits, range 6000.00–40000.00

Domain defines the **data type** and **the set of allowable values** for one or more attributes.

E.g. Dates have various formats such as yyyy-mm-dd, or dd mm,yyyy etc.

DEFINITION SUMMARY

Informal Terms	Formal Terms
Table	Relation
Column	Attribute
Row	Tuple
Values in a column	Domain
Table Definition	Schema of a Relation

Properties of Relations

- Relation name is distinct from all other relation names in relational schema.
- Each cell of relation contains exactly one atomic (single) value.
- Each attribute within a relation has a distinct name.
- Values of an attribute are all from the same domain.
- Each record is distinct; there are no duplicate records.
- Order of attributes has no significance.
- Order of records has no significance, theoretically.

Candidate Key (CK)

- Candidate Key (CK)
  - An attribute, or smallest set of attributes, that uniquely identifies a record within a relation.
  - There may be several candidate keys for a relation
  - A key with more than one attribute is a **composite** key.

Examples on Candidate Key

Consider the Branch relation in Fig. 4.1

- Given a value of city, we can determine several branch offices (e.g. London has 2 branch offices)
- Hence, city cannot be a candidate key.
- Given a branch number value, branchNo, we can determine at most one record.
- So, branchNo is a candidate key.

Examples on Candidate Key

Consider the Viewing relation in Fig. 4.3

clientNo	propertyNo	viewDate	Comment
CR56	PA14	24-May-08	Too small
CR76	PG4	20-Apr-08	Too remote
CR56	PG4	26-May-08	
CR62	PA14	14-May-08	No dining room
CR56	PG36	28-Apr-08	

- Given a clientNo, there may be several viewings for different properties.
- Given a propertyNo, there may be several clients who viewed this property.
- Therefore, clientNo by itself or propertyNo by itself cannot be selected as a candidate key.
- However, combination of clientNo and propertyNo identifies at most one record for this instance.
- So, together they form the (composite) candidate key.
- **How about a client may view a property more than once?**

Examples on Candidate Key

Consider the Viewing relation in Fig. 4.3

clientNo	propertyNo	viewDate	Comment
CR56	PA14	24-May-08	Too small
CR76	PG4	20-Apr-08	Too remote
CR56	PG4	26-May-08	
CR62	PA14	14-May-08	No dining room
CR56	PG36	28-Apr-08	

- **If a client may view a property more than once:**
  - **Given that your design already has (clientNo, propertyNo) as the Primary Key, the operation of adding CR56 viewing PA14 will FAIL!!!!**
  - **In this case, ask if updating viewDate to hold the latest viewDate is feasible to solve the issue.**
  - **If NOT accepted, that means your design has to be changed to use (clientNo, propertyNo, viewDate) as the composite primary key.**

Examples on Candidate Key

Consider the Viewing relation in Fig. 4.3

clientNo	propertyNo	viewDate	Comment
CR56	PA14	24-May-08	Too small
CR76	PG4	20-Apr-08	Too remote
CR56	PG4	26-May-08	
CR62	PA14	14-May-08	No dining room
CR56	PG36	28-Apr-08	

- So, different operations will result in a different selection of candidate key.
- For Dream Home case study, the choice of the textbook is **(clientNo, propertyNo).**
- Another possible design is to create a Viewing Form number as the primary key instead.

Primary Key (PK)

- Primary Key (PK)
  - Candidate key selected to identify records uniquely within relation.
  - In the worst case, the entire set of attributes could serve as the primary key.
  - Every relation must have a primary key.

Choices for Primary Key

In choosing the column(s) to be used as primary key, consider these properties:

- Values of primary key shall be unique, i.e. no duplicate value.
- Primary key shall always have a value. In other words, it shall not contain NULL.
- The value of primary key should not change.
  Primary key is used to reference other tables. If you change its value, you have to change all its references; otherwise, the references will be lost.

Choices for Primary Key (con't)

- Primary key is usually a single column (e.g., customerID or productCode). But some circumstances, it could be made up of several columns. You should use as few columns as possible.
- Primary key often uses integer (or number) type. It could also be other types, such as texts. However, it is best to use numeric column as primary key for efficiency.

Surrogate Key

- A surrogate key is an artificially generated key.
- Useful when your records essentially have no natural key.
- Most often, implemented as integers in an automatically incrementing field.
- The main advantage of the surrogate key is that they're easy to guarantee as unique.
- The main disadvantage is that they don't have any meaning.

Surrogate key - disadvantage

- Primary key could take an arbitrary number. Most RDBMSs support so-called *auto-increment* (or AutoNumber type) for integer primary key, where (current maximum value + 1) is assigned to the new record. *However, in some circumstances, it is not desirable where a composite primary key is better to avoid inconsistency of data.*
  - E.g. We have a CustomerProducts relation that relates customers to products with an "orderLimit" attribute to state the amount of product each customer is allowed to order.

    CustomerProducts (custID, productID, orderLimit)

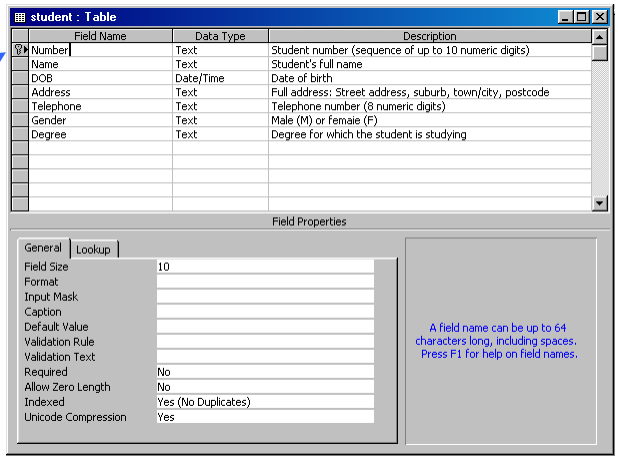
    If you use an autonumber as the primary key, you might run into the following situation:
CustomerProductID	CustomerID	ProductID	OrderLimit
1	1	100	25
2	1	100	30
  - What is the problem here? The primary issue is data integrity.

Alternate Keys

- Alternate Keys
  - Candidate keys that are not selected to be primary key.

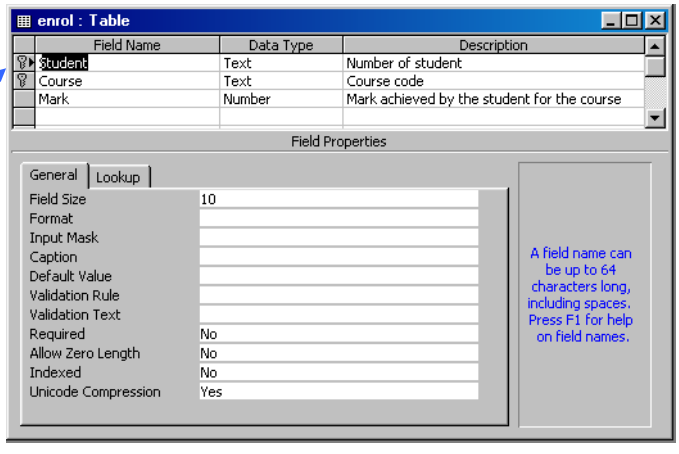
Primary Key Example in MS Access

Primary key (table in "design view" in MsAccess)



Composite Primary Key Example in MS Access

Composite primary key

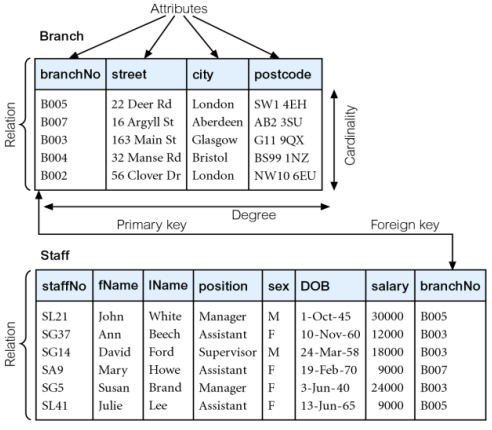


Foreign Key (FK)

- A foreign key (FK) is a “copy” of a candidate key that has been exported from one table and added as a new column in another table to represent the relationship between them.
- A foreign key is a copy of the whole of its parent candidate key
  - if the primary key is composite then so is the foreign key
- A foreign key is needed to obtain the details from the parent table which are not duplicated in the child table because **data duplication might result in data inconsistency**.
- The foreign key is the minimal amount of data redundancy that is needed.

Foreign Key (FK)

- For example, the inclusion of branchNo in both the Branch and Staff relations. (see Fig. 4.1)



Foreign Key Example

Textbook

Title	ISBN	Edition	Publisher ID
Database Systems	0-201-34287-1	2	P091
Database Processing	0-02-366881-4	5	P473
Analysis and Design	0-202-36995-4	null	P091

Foreign Key

Primary Key

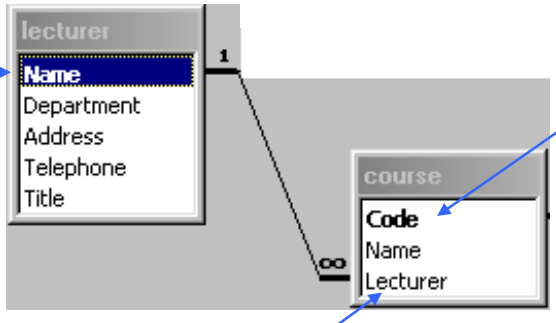
Publisher

Publisher ID	Name
P091	Longman
P473	University Press

Primary Key

Foreign Key Example in MS Access

Primary key of Lecturer table



Primary key of Course table

Foreign key

Note: The name of the FK attribute does not need to match the PK – but the domain should be the same.

Representing Relational Database Schemas

- A common convention for representing a relation schema is to give the name of the relation following by the attribute names in parentheses.
- Normally, the primary key is underlined.

  Branch (branchNo, street, city, postcode)
- The *conceptual model*, or *conceptual schema*, is the set of all such schemas for the database.
- The relational schema for part of the DreamHome case study is as follows:

Branch	(branchNo, street, city, postcode)
Staff	(staffNo, fName, lName, position, sex, DOB, salary, branchNo)
PropertyForRent	(propertyNo, street, city, postcode, type, rooms, rent, ownerNo, staffNo, branchNo)
Client	(clientNo, fName, lName, telNo, prefType, maxRent, eMail)
PrivateOwner	(ownerNo, fName, lName, address, telNo, eMail, password)
Viewing	(clientNo, propertyNo, viewDate, comment)
Registration	(clientNo, branchNo, staffNo, dateJoined)

An instance of all such schemas for DreamHome case study (Fig. 4.3)

Branch

branchNo	street	city	postcode
B005	22 Deer Rd	London	SW1 4EH
B007	16 Argyll St	Aberdeen	AB2 3SU
B003	163 Main St	Glasgow	G11 9QX
B004	32 Manse Rd	Bristol	BS99 1NZ
B002	56 Clover Dr	London	NW10 6EU

Staff

staffNo	fName	lName	position	sex	DOB	salary	branchNo
SL21	John	White	Manager	M	1-Oct-45	30000	B005
SG37	Ann	Beech	Assistant	F	10-Nov-60	12000	B003
SG14	David	Ford	Supervisor	M	24-Mar-58	18000	B003
SA9	Mary	Howe	Assistant	F	19-Feb-70	9000	B007
SG5	Susan	Brand	Manager	F	3-Jun-40	24000	B003
SL41	Julie	Lee	Assistant	F	13-Jun-65	9000	B005

PropertyForRent

propertyNo	street	city	postcode	type	rooms	rent	ownerNo	staffNo	branchNo
PA14	16 Holhead	Aberdeen	AB7 5SU	House	6	650	CO46	SA9	B007
PL94	6 Argyll St	London	NW2	Flat	4	400	CO87	SL41	B005
PG4	6 Lawrence St	Glasgow	G11 9QX	Flat	3	350	CO40		B003
PG36	2 Manor Rd	Glasgow	G32 4QX	Flat	3	375	CO93	SG37	B003
PG21	18 Dale Rd	Glasgow	G12	House	5	600	CO87	SG37	B003
PG16	5 Novar Dr	Glasgow	G12 9AX	Flat	4	450	CO93	SG14	B003

Client

clientNo	fName	lName	telNo	prefType	maxRent	eMail
CR76	John	Kay	0207-774-5632	Flat	425	john.kay@gmail.com
CR56	Aline	Stewart	0141-848-1825	Flat	350	astewart@hotmail.com
CR74	Mike	Ritchie	01475-392178	House	750	mritchie01@yahoo.co.uk
CR62	Mary	Tregear	01224-196720	Flat	600	maryt@hotmail.co.uk

PrivateOwner

ownerNo	fName	lName	address	telNo	eMail	password
CO46	Joe	Keogh	2 Fergus Dr, Aberdeen AB2 7SX	01224-861212	jkeogh@lhh.com	********
CO87	Carol	Farrel	6 Achray St, Glasgow G32 9DX	0141-357-7419	cfarrel@gmail.com	********
CO40	Tina	Murphy	63 Well St, Glasgow G42	0141-943-1728	tinam@hotmail.com	********
CO93	Tony	Shaw	12 Park Pl, Glasgow G4 0QR	0141-225-7025	tony.shaw@ark.com	********

Viewing

clientNo	propertyNo	viewDate	comment
CR56	PA14	24-May-08	too small
CR76	PG4	20-Apr-08	too remote
CR56	PG4	26-May-08	
CR62	PA14	14-May-08	no dining room
CR56	PG36	28-Apr-08	

Registration

clientNo	branchNo	staffNo	dateJoined
CR76	B005	SL41	2-Jan-08
CR56	B003	SG37	11-Apr-07
CR74	B003	SG37	16-Nov-06
CR62	B007	SA9	7-Mar-07

Integrity Constraints

- Integrity constraints ensure that data is accurate.
- Domain constraints
  - form restrictions on the data type and set of values allowed for the attributes of relations.
- Two important integrity rules:
  - Entity integrity
  - Referential integrity

Entity integrity

- Entity Integrity: In a base relation, no attribute of a primary key can be null.
  - Null
    - Represents value for an attribute that is currently unknown or not applicable for record.
    - Deals with incomplete or exceptional data.
    - Represents the absence of a value

Referential integrity

- Referential Integrity
  - If foreign key exists in a relation, either foreign key value must match a candidate key value of some record in its home relation or foreign key value must be wholly null

Referential integrity

- Staff oversees PropertyForRent

Staff

PK

Relation

staffNo	fName	lName	position	sex	DOB	salary	branchNo
SL21	John	White	Manager	M	1-Oct-45	30000	B005
SG37	Ann	Beech	Assistant	F	10-Nov-60	12000	B003
SG14	David	Ford	Supervisor	M	24-Mar-58	18000	B003
SA9	Mary	Howe	Assistant	F	19-Feb-70	9000	B007
SG5	Susan	Brand	Manager	F	3-Jun-40	24000	B003
SL41	Julie	Lee	Assistant	F	13-Jun-65	9000	B005

FK

PropertyForRent

PK

propertyNo	street	city	postcode	type	rooms	rent	ownerNo	staffNo	branchNo
PA14	16 Holhead	Aberdeen	AB7 5SU	House	6	650	CO46	SA9	B007
PL94	6 Argyll St	London	NW2	Flat	4	400	CO87	SL41	B005
PG4	6 Lawrence St	Glasgow	G11 9QX	Flat	3	350	CO40		B003
PG36	2 Manor Rd	Glasgow	G32 4QX	Flat	3	375	CO93	SG37	B003
PG21	18 Dale Rd	Glasgow	G12	House	5	600	CO87	SG37	B003
PG16	5 Novar Dr	Glasgow	G12 9AX	Flat	4	450	CO93	SG14	B003

Referential Integrity Example

Course

Foreign Key

Name	Level	Credits	Lecturer ID
Database Systems	Undergrad	3	38421
Data Networks	Undergrad	6	null
Speech Processing	Postgrad	null	34561

Violates referential integrity

Note:

Foreign Key attribute name not necessarily the same as primary key

Primary Key

Lecturer

Staff ID	StaffName
91027	Goscinski
38421	Nguyen

Why do NoSQL systems emerge in the 2000s?

- NoSQL systems relax the rigidity of storing data in tables by allowing a diverse set of data types.
- **NoSQL database** technology is a database type that stores information in JSON documents instead of columns and rows used by relational databases.
- Consequently, NoSQL databases are built to be flexible, scalable, and capable of rapidly responding to the data management demands of modern businesses.
- They allow for faster initial application development.
- However, NoSQL systems lack traditional systems' support for strong data consistency, instead relying on a weaker concept of eventual consistency.

NoSQL systems: drawbacks

- NoSQL databases lack native, built-in referential integrity (like foreign keys) to enforce relationships.
- In a relational model, data would be stored in several tables, providing redundancy and enforcement. This means the relational model comes with a built-in, foolproof way of ensuring business logic and trustworthiness at the database layer.
  - This is why relational databases continue to be popular.
- However, the highest priority for web-based applications is the ability to service large numbers of user requests, which is the strength of NoSQL databases.

Summary

We have covered the following:

- **Relational model concepts**
  - Relations
  - Attributes
  - Relationship
- **Integrity constraints**
  - Entity integrity
  - Referential integrity
- **Terms:**
  - **Domain**
  - **Candidate key, Primary key and Foreign Key**